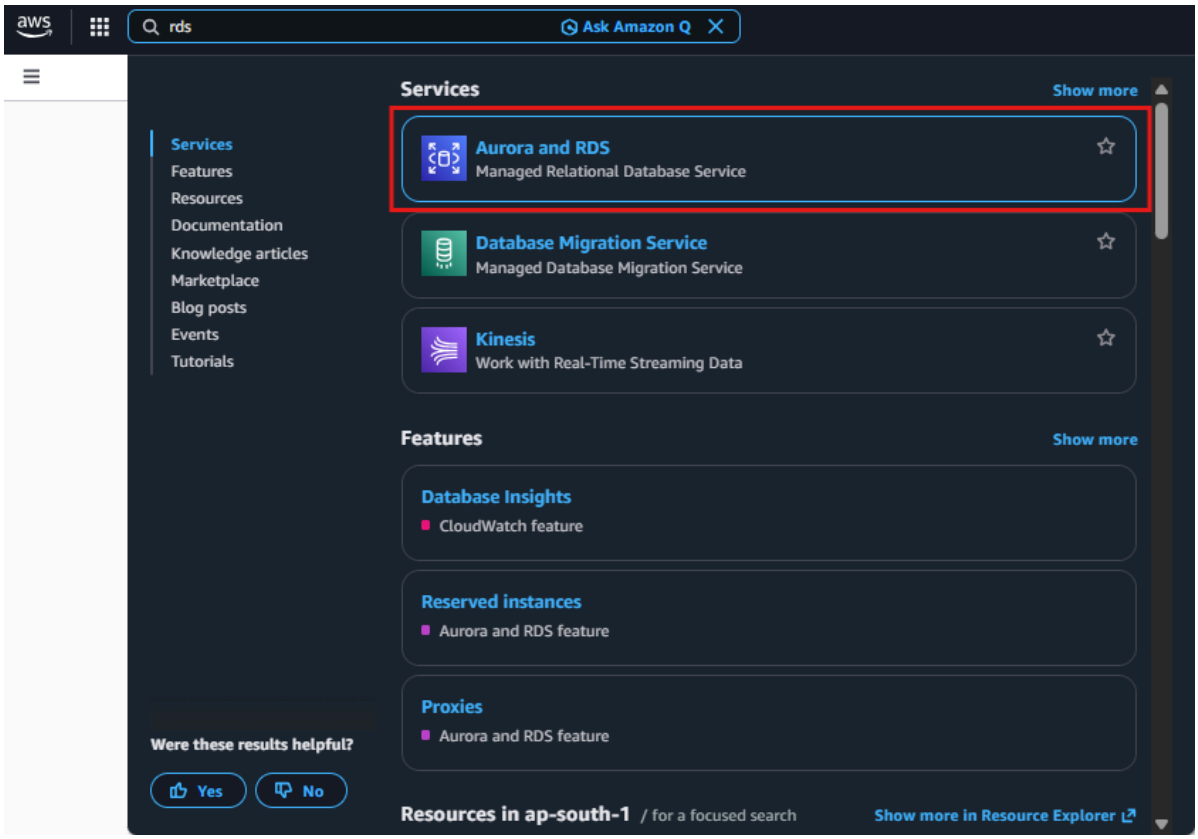
Lab: Setup Amazon Aurora MySQL and Load Rental Data

Objective

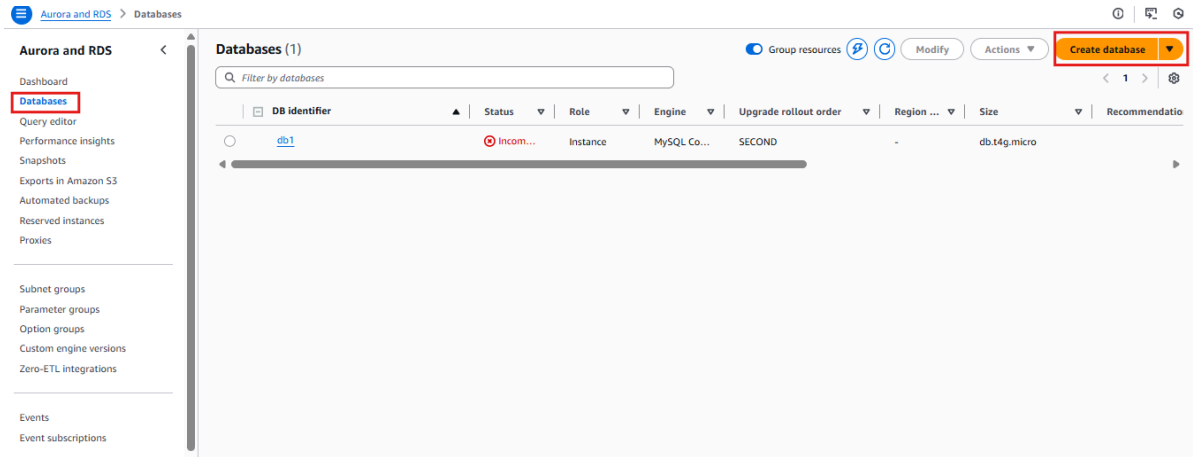
Provision an Amazon Aurora MySQL database on AWS, configure network access, connect from a local machine, create tables, and load CSV data into the database.

Step 1: Create Amazon Aurora MySQL Database

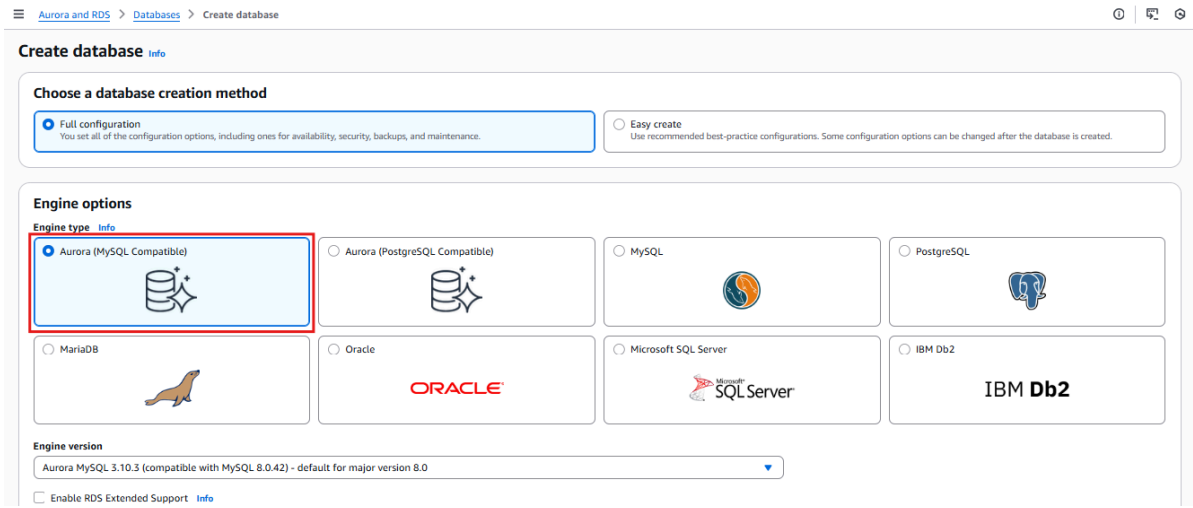
1. Log in to the AWS Console and search for **RDS**.



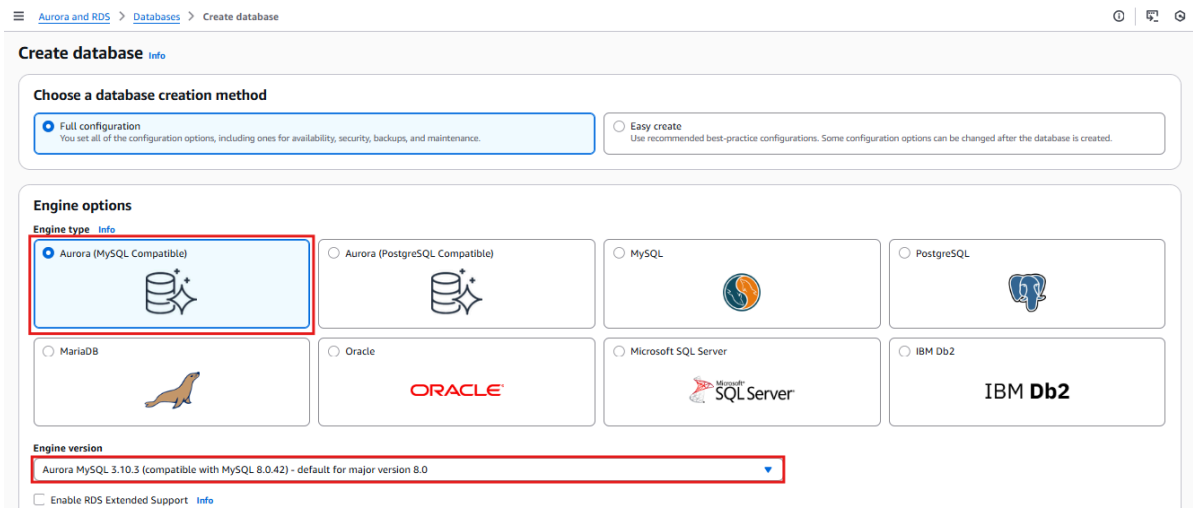
2. Click **Databases** and then select **Create database**.



3. Under **Engine options**, choose **Amazon Aurora MySQL-Compatible Edition**.

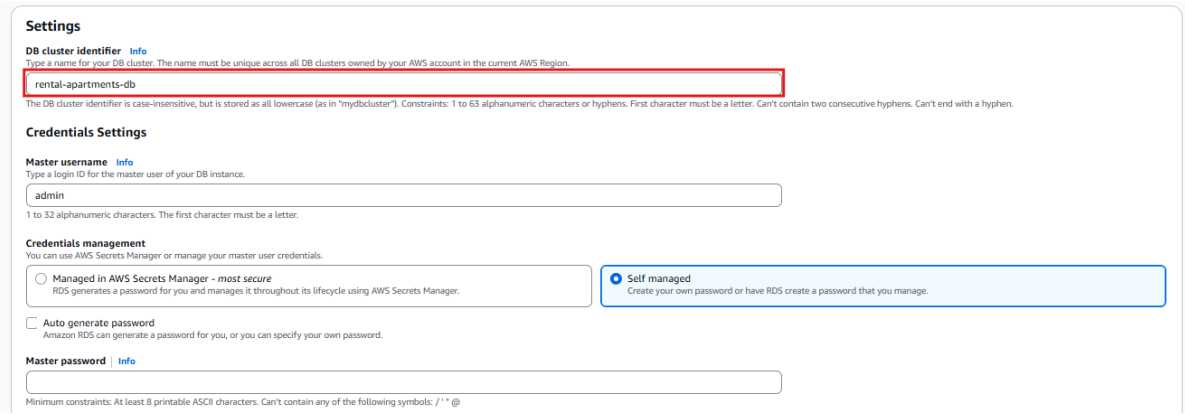


4. Keep the default version (Aurora MySQL 3.05).

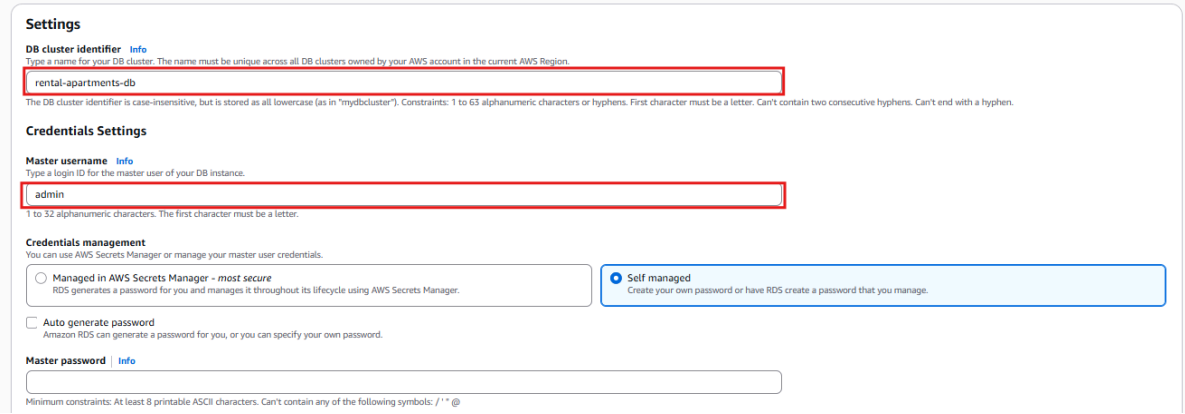


5. Under **Templates**, select **Dev/Test**.

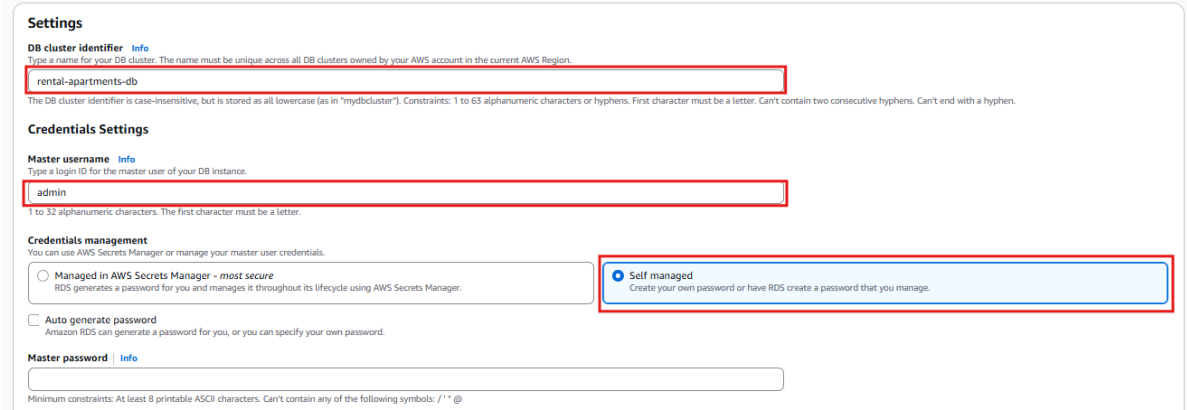
6. Provide a database identifier, for example: `rental-apartments-db`.



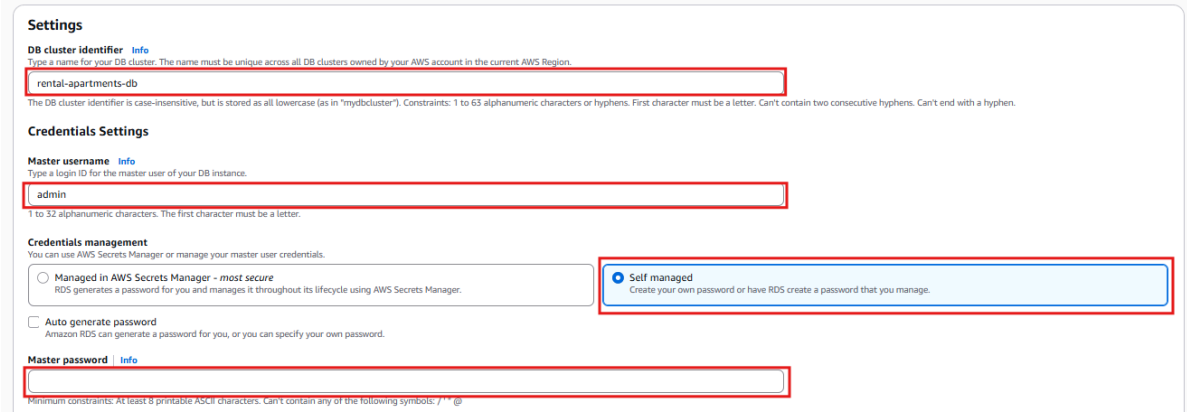
7. Set the master username as `admin`.



8. Choose **Self-managed credentials**.

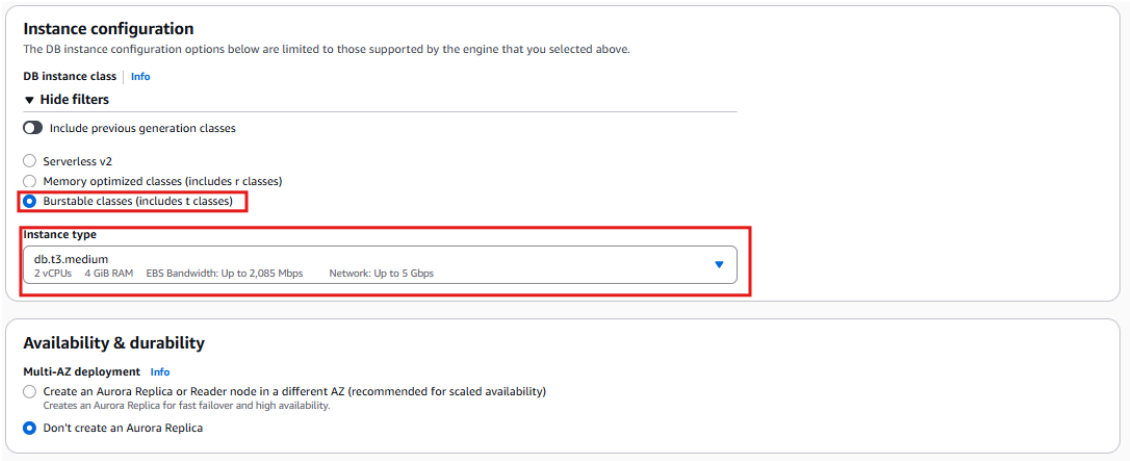


9. Enter and confirm a password (make a note of it).



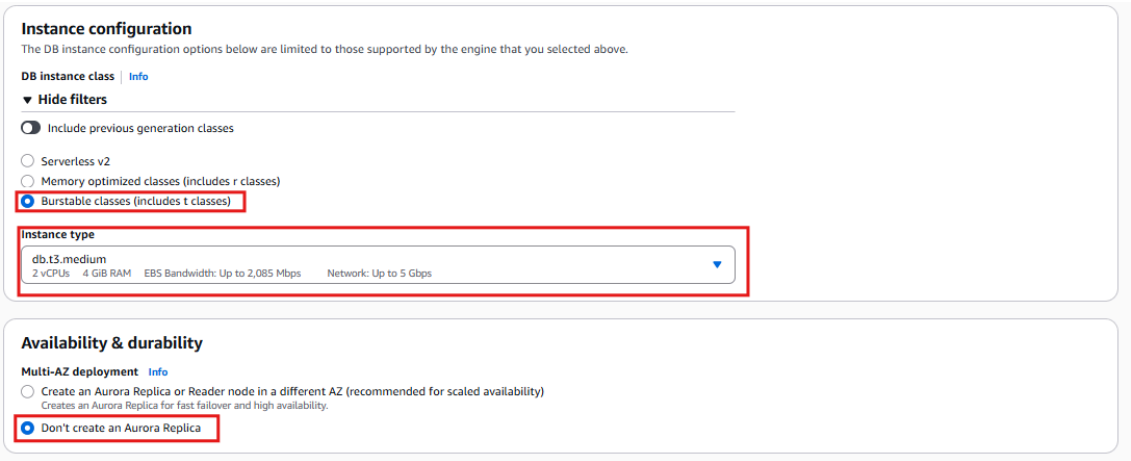
10. Under **DB instance class**, select:

    - **Burstable classes**
    - Choose **db.t3.medium** (or Serverless v2 if preferred)



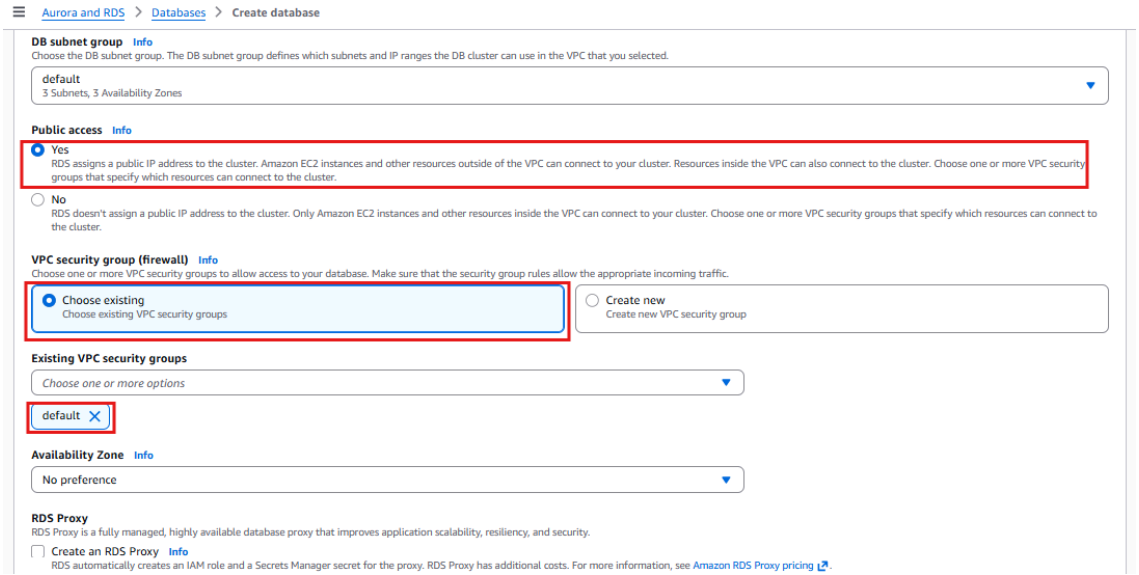
11. Under **Availability & durability**, select:

    - Do not create a replica

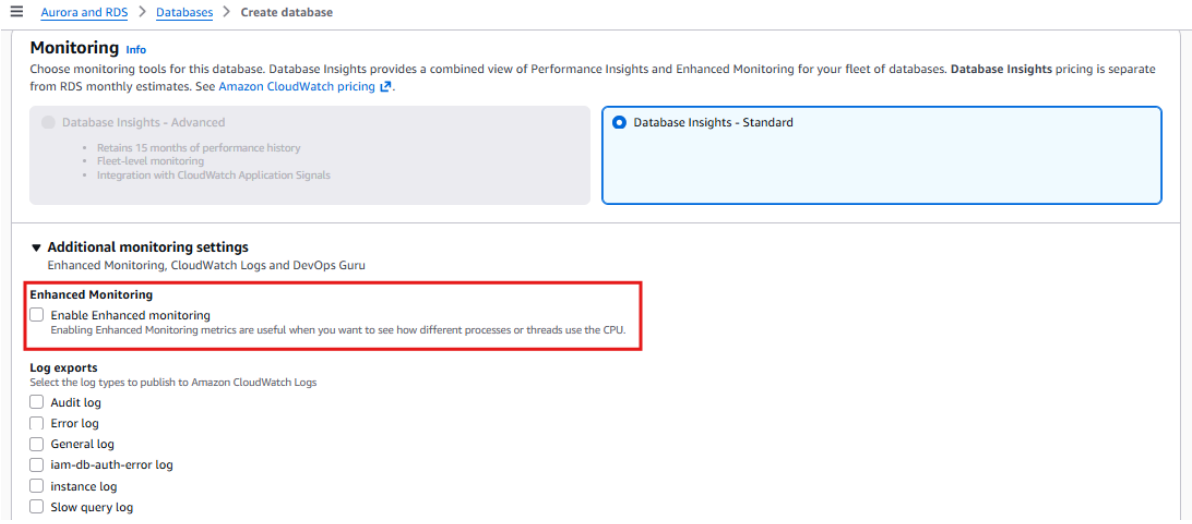


12. Under **Connectivity**, set:

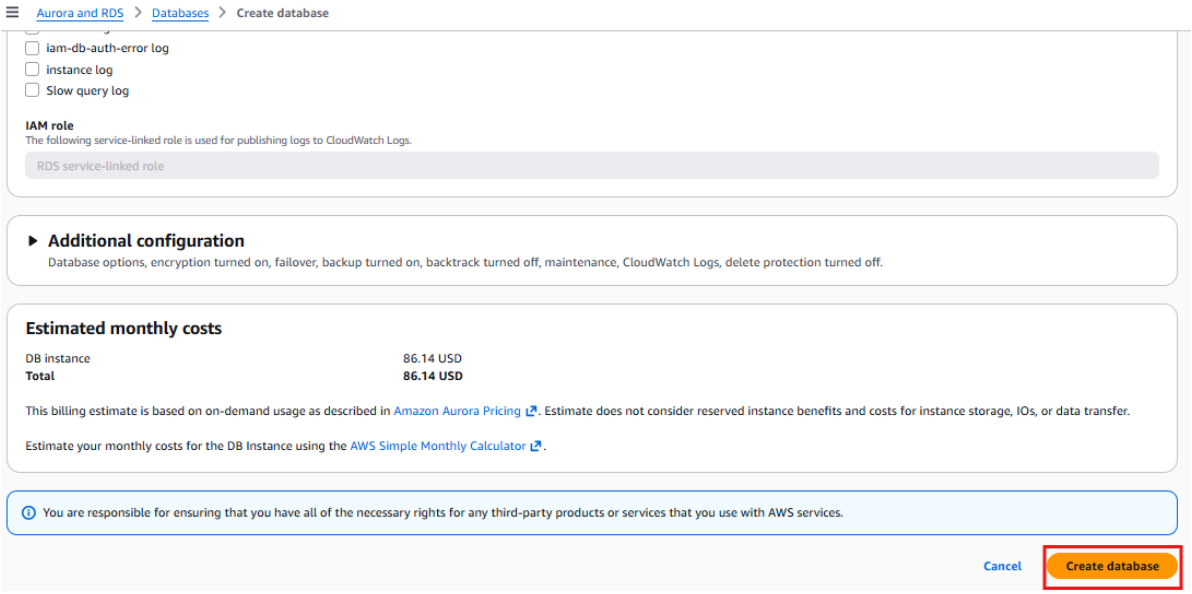
    - Public access → **Yes**
    - Use existing VPC security group (default)



13. Uncheck **Enhanced monitoring**.



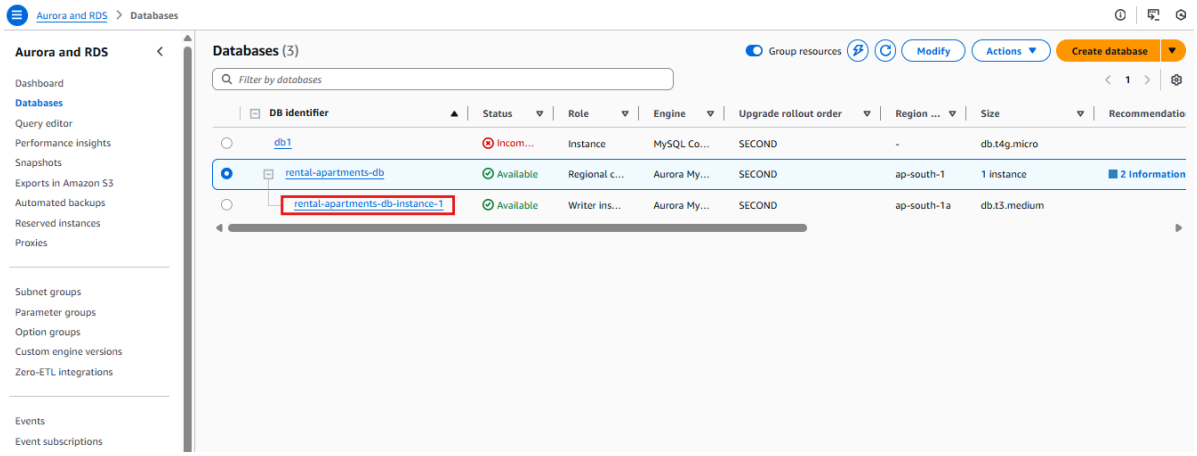
14. Click **Create database**.



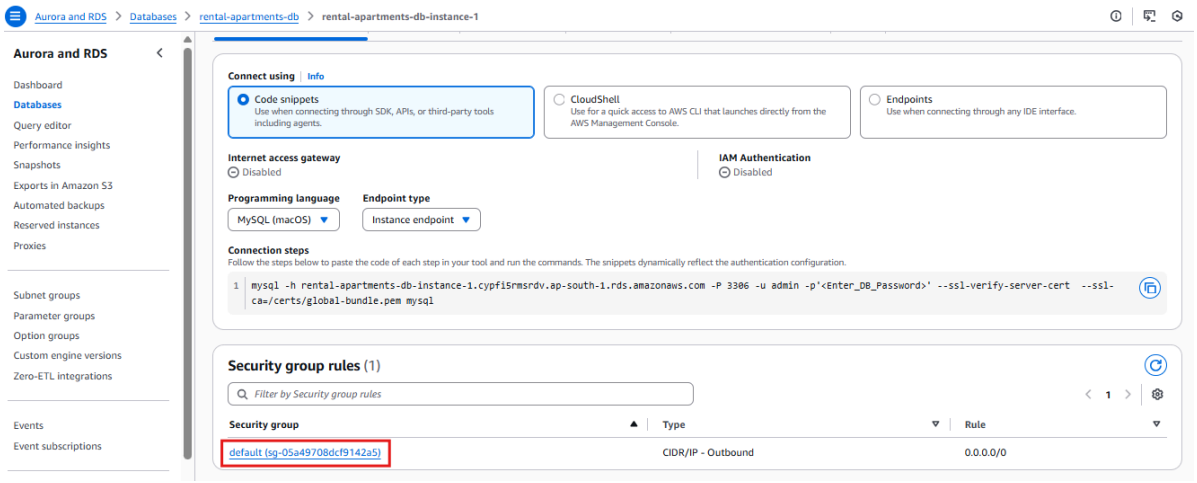
Wait approximately 10 minutes for the database to be fully provisioned.

Step 2: Configure Security Group for MySQL Access

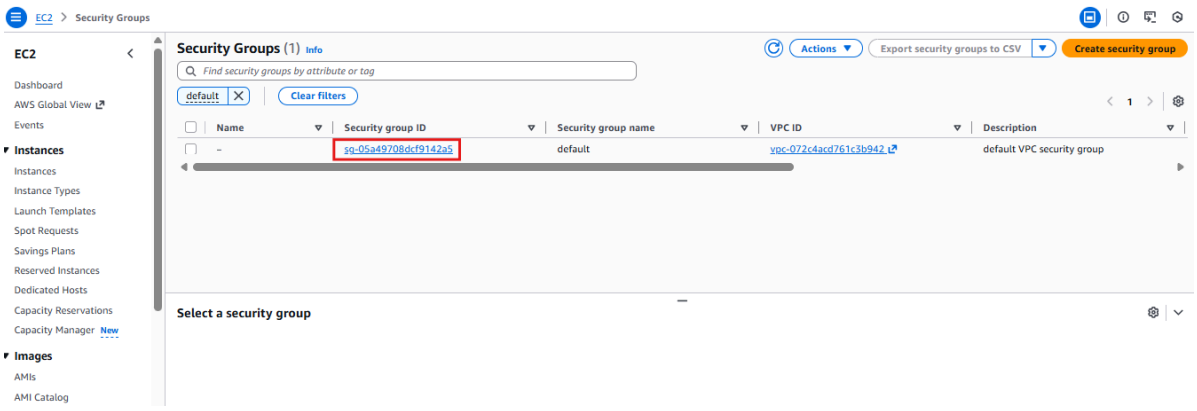
1. Once the database is created, click on the database identifier.



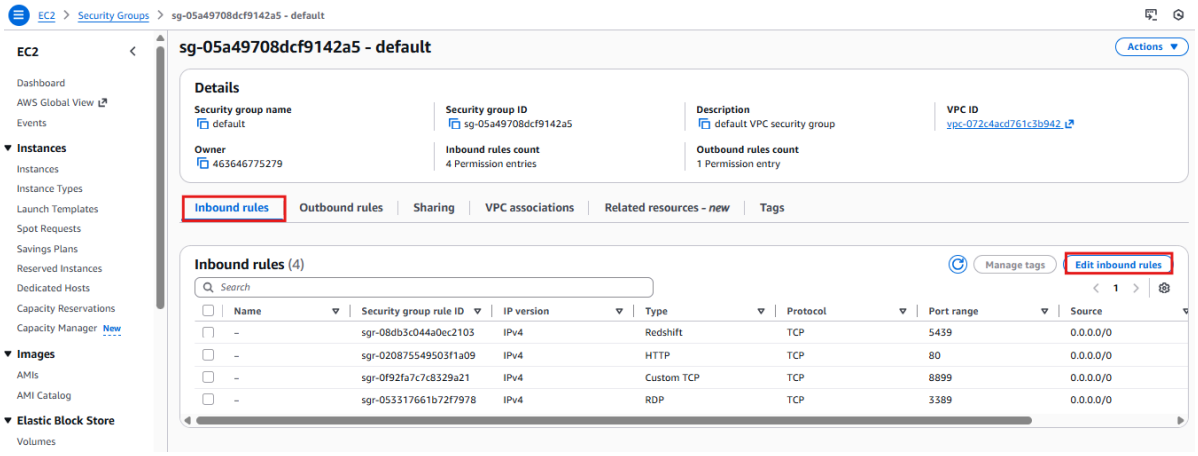
2. Under **Connectivity & security**, locate the **VPC security group**.



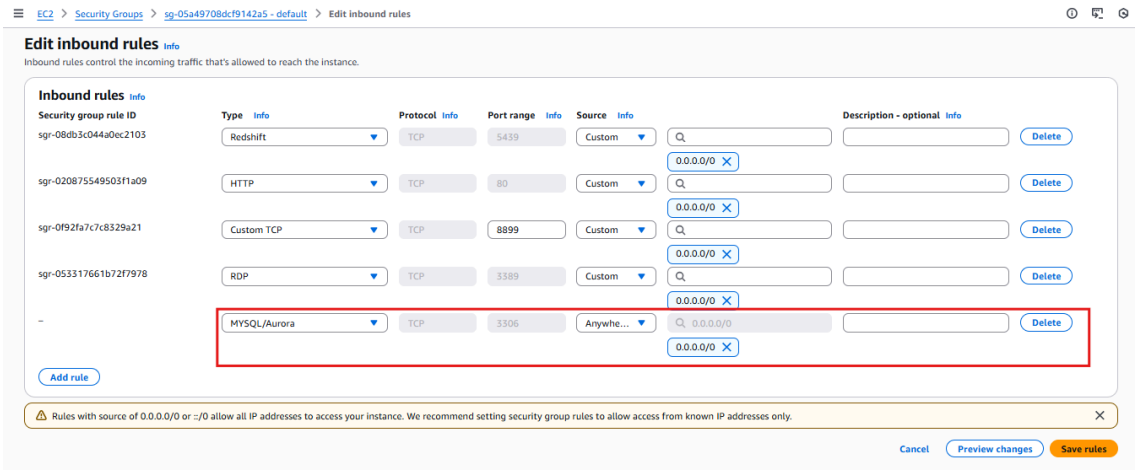
3. Click the security group link to open it.



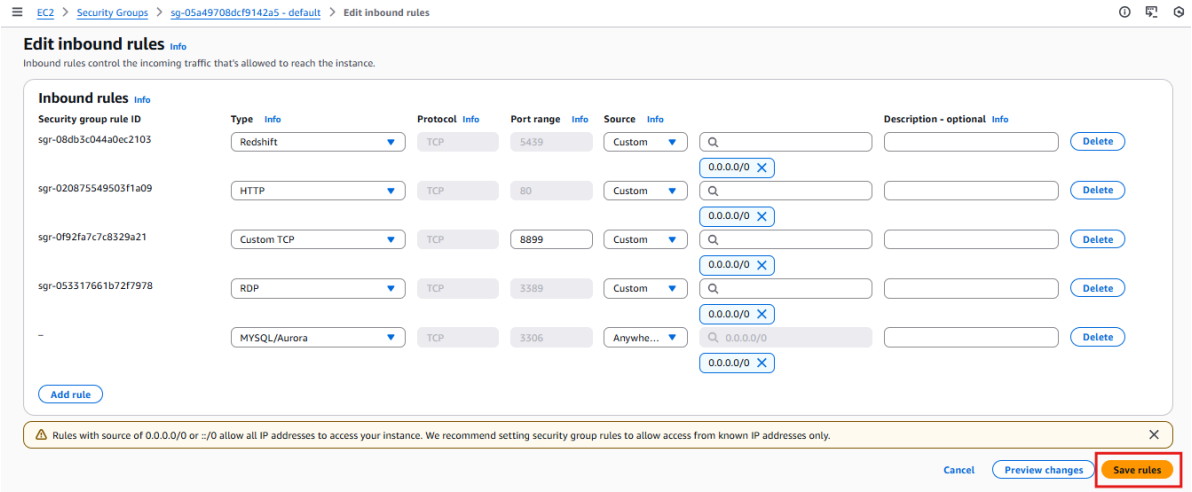
4. Go to **Inbound rules** and click **Edit inbound rules**.



5. Click Add a new rule:
   - Type → **MySQL/Aurora**
   - Port → **3306** (auto-filled)
   - Source → **0.0.0.0/0** (allow all IPs for lab purposes)



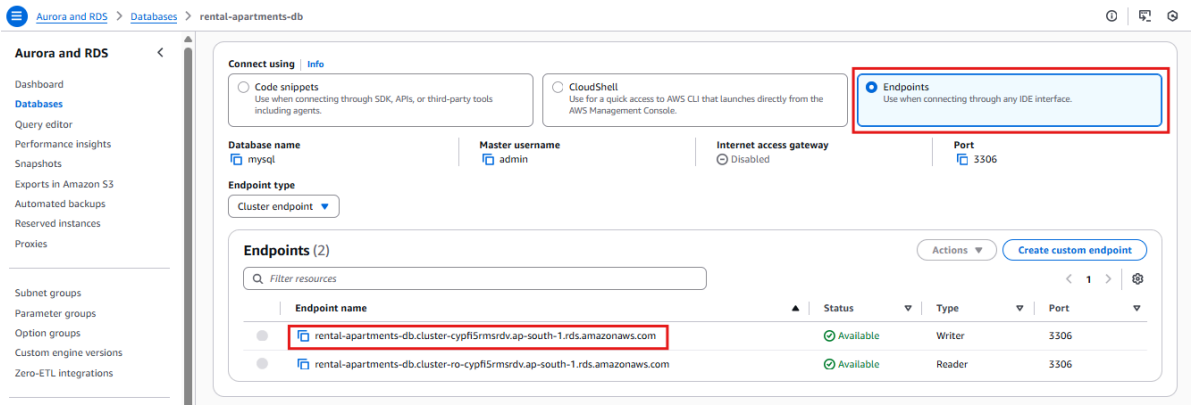
6. Click **Save rules**.



Note: Allowing 0.0.0.0/0 is not recommended in production but acceptable for lab testing.

Step 3: Connect to Aurora from Local Machine

1. Copy the **Cluster endpoint** from the RDS console.



2. Ensure MySQL is installed locally.
3. Verify installation using:

```
mysql --version
```

4. Navigate to the working directory containing your SQL files.
5. Connect using:

```
mysql -h <AURORA-ENDPOINT> -u admin -p
```

6. Enter the password created earlier.

You are now connected to the Aurora MySQL instance.

Step 4: Create Database and Tables

1. Create the database:

```
CREATE DATABASE rental_apartments;
```

2. Use the database:

```
USE rental_apartments;
```

3. Execute the DDL scripts to create the following tables:
   - apartments
   - apartment_attributes
   - apartment_viewings
4. Verify table creation:

```
SHOW TABLES;
```

You should see all three tables listed.

Step 5: Load CSV Data into Tables

1. Use the `LOAD DATA LOCAL INFILE` command to load data from local CSV files:

```
LOAD DATA LOCAL INFILE 'data/apartments.csv'
INTO TABLE rental_apartments.apartments
FIELDS TERMINATED BY ','
IGNORE 1 ROWS;
```

2. Repeat for:
    - apartment_attributes.csv
    - apartment_viewings.csv
3. Verify row ingestion from MySQL output.
    - apartments → ~10,000 rows
    - apartment_attributes → ~10,000 rows
    - apartment_viewings → ~5,000 rows

Outcome

An Amazon Aurora MySQL database was successfully provisioned.

Local connectivity was established via MySQL client.

Three relational tables were created and populated with CSV data, completing the initial database setup for the rental apartments application.